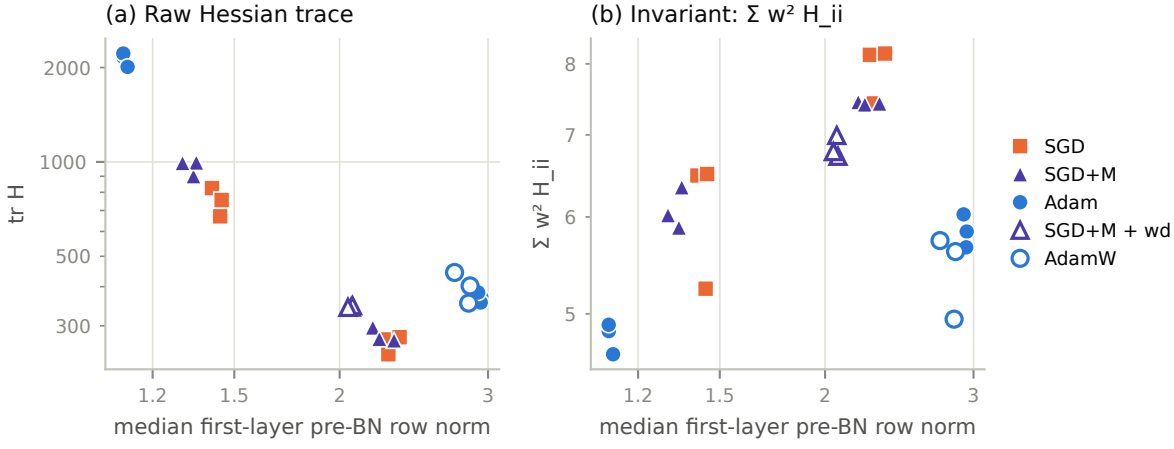
(a) Raw Hessian trace
tr H
2000
1000
500
300
1.2
1.5
2
3
median first-layer pre-BN row norm
(b) Invariant: Σ w² H_ii
Σ w² H_ii
8
7
6
5
1.2
1.5
2
3
median first-layer pre-BN row norm
SGD
SGD+M
Adam
SGD+M + wd
AdamW